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PROJECT AND TEAM INFORMATION

Project Title

Smart Courier Delivery and Route Optimization System

Team Name: *Team #*	*Route Rangers* *DSCPP-III-2026-T332* *Section: J and ML4*
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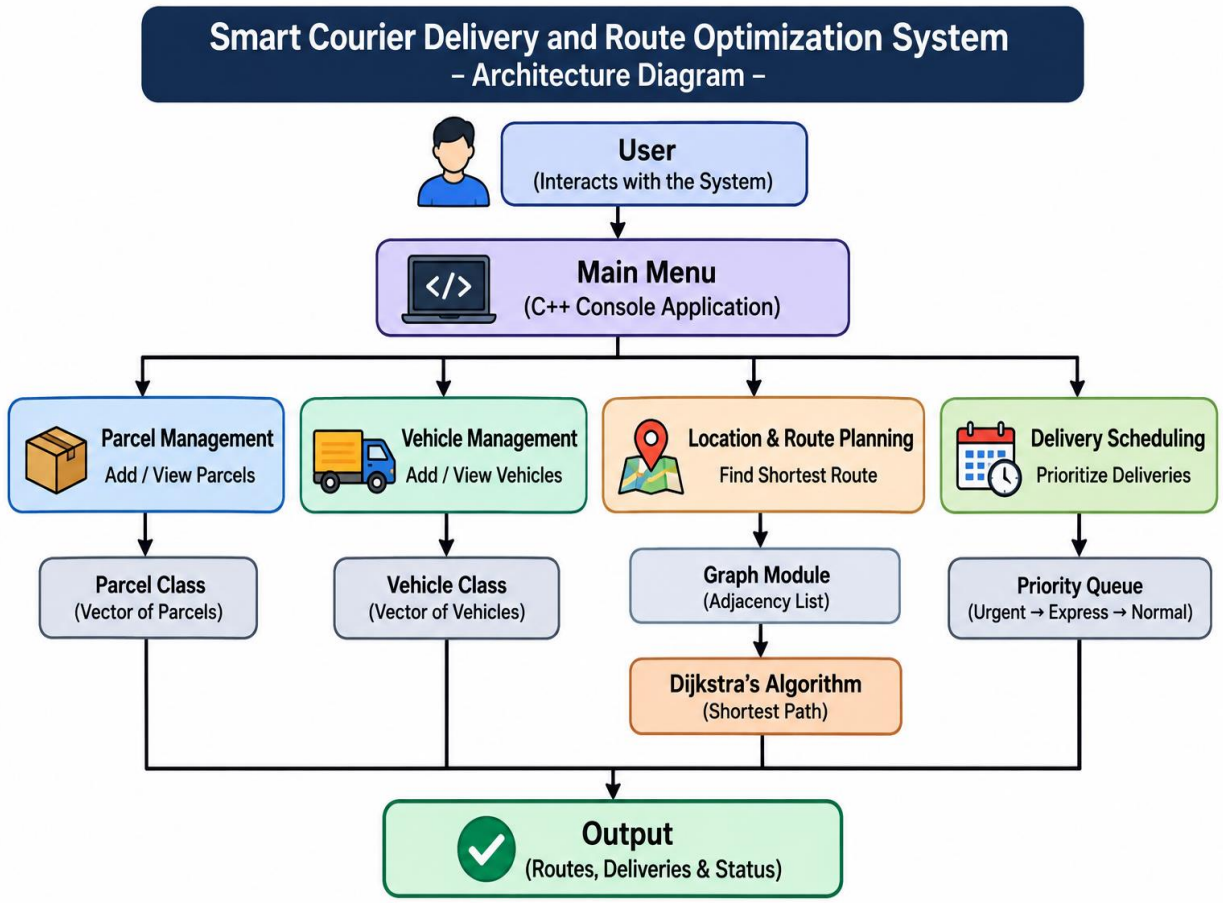
PHASE-III PROJECT PROGRESS DESCRIPTION

Project Abstract

Our project is a C++ console-based courier management prototype. It stores parcel and vehicle details, represents delivery locations and roads as a weighted graph, finds a shortest route between selected locations using Dijkstra's algorithm, and processes delivery tasks according to priority. The aim is to bring basic parcel management, route finding, and priority scheduling together in one program using object-oriented programming and data structures.

Updated Project Approach and Architecture

The program is organized into separate modules and accessed through a menu-driven interface. The Parcel and Vehicle classes keep related information together, while vectors store the records during the program session. The road network is represented by an adjacency-list graph: locations are vertices and roads are weighted edges. Dijkstra's algorithm uses a priority queue to calculate a shortest path for non-negative road distances. A separate priority queue orders delivery tasks so urgent tasks are processed before express and normal tasks. The implementation uses standard C++ libraries such as iostream, string, vector, and queue/priority_queue. It currently runs as a console application; records are held in saved files. No network communication protocol is used.



Tasks Completed

Task Completed	Team Member
Parcel and vehicle classes; add/view operations and menu integration.	Arpit Rai
Graph representation; location and road management.	Shreshth Vardhan
Dijkstra shortest-path calculation and route display.	Shreshth Vardhan
Priority-queue delivery scheduling and processing.	Shreshth Vardhan
Integration and testing of core menu options.	Both members

Challenges/Roadblocks

During development, we had to connect separate modules and keep the menu, parcel/vehicle records, graph, and delivery queue working together. Route testing also required checking that locations and roads were entered correctly and that road distances were positive. The current prototype does not yet prevent all invalid parcel or vehicle inputs, check duplicate IDs, enforce vehicle capacity, or save records after the program closes. We plan to address these points through additional input checks, duplicate-ID checks, capacity rules, and file handling. We will continue testing with small known graphs and different delivery priorities before considering advanced multi-stop optimization.

Project Outcome/Deliverables

The current deliverable is a working C++ console prototype with parcel and vehicle management, location and road management, shortest-route calculation using Dijkstra's algorithm, and priority-based delivery processing. The project also includes the source code and test evidence for the implemented features. File storage, vehicle-capacity enforcement, and advanced route optimization are also implemented on our project.

Progress Overview

The main program structure and core modules have been implemented and tested: parcel/vehicle management, graph construction, Dijkstra route finding, and priority-queue scheduling. These core features are on track for the current phase. Input validation beyond the existing road checks, duplicate-ID handling, vehicle-capacity checks, and permanent file storage are also done. The remaining effort is focused on improving reliability, testing edge cases, and preparing clear documentation and screenshots.

Codebase Information

Repository: https://github.com/Anonymous0-27/Smart-Courier-Route-Optimizer.git Main branch: main
The repository contains the C++ project source and README. The work is divided across parcel/vehicle management, graph and Dijkstra routing, and priority scheduling.

Testing and Validation Status

Test Type	Status (Pass/Fail)	Notes
Parcel/vehicle add and view	Pass	Valid sample records displayed.
Shortest route (Dijkstra)	Pass	Test graph returned the route.
Priority delivery queue	Pass	Urgent tasks processed before.
Invalid weight/capacity checks	Pass	Comprehensive validation implemented.

Deliverables Progress

Completed: Parcel and Vehicle classes with add/view operations; menu-driven integration; location and road management; graph representation; Dijkstra shortest-route calculation; and priority-queue delivery scheduling. Testing of the core features has been carried out, broader input validation, duplicate-ID checks, vehicle-capacity enforcement, and more edge-case tests has been done. File handling is completed. The final demonstration will show how vectors, queues, priority queues, graphs, searching/sorting where applicable, file handling, classes and constructors are used to solve a practical problem.